

M004 · Functions with Parameters & Returns — English Worksheet

Topic: Functions with Parameters & Returns · **Course:** Functions & Games · **Time:** about 45 minutes

This lesson you give a function a **parameter** — a number it uses inside, so calling it with different numbers builds different things. A function can also **give back** an answer.

1 · Predict 🧠

```
define build_tower(height):  
    repeat height times:  
        place block  
        move up  
  
build_tower(3)
```

The function uses **height** inside the repeat. How many blocks tall is this tower?


```
define build_tower(height):  
    repeat height times:  
        place block  
        move up  
  
build_tower(2)  
build_tower(5)
```

The same function is called twice with different numbers. What gets built? Are the two towers the same?


```
def count_steps(distance):  
    give back distance + 2  
  
steps = count_steps(4)  
say steps
```

This function gives an answer back instead of building. What number does the agent say?

2 · Spot the Bug 🐛

Each block below is broken. Rewrite it so it works, then explain why the original was wrong and why your fix works.

Bug A — The code should build a tower 4 blocks tall. But when you run it, **nothing happens at all**.

```
define build_tower(height):  
    repeat height times:  
        place block  
        move up
```

Hint: defining a function only **teaches** it. Something is missing after the definition.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — The function should build a tower as tall as the number you give it. But `build_tower(7)` and `build_tower(2)` both build the **same** tower.

```
define build_tower(height):  
    repeat 3 times:  
        place block  
        move up  
  
build_tower(7)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The function needs a number to know how tall to build. But the call forgot to give it one, so the code breaks.

```
define build_tower(height):  
    repeat height times:  
        place block  
        move up  
  
build_tower()
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a tower village: write **one** function with a **height** parameter, then call it **three** times with different numbers.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.